



## Trigonometric Equations: RACE-01

## Single Correct Question

1.  $\tan(p\pi/4) = \cot(q\pi/4)$  if:  
 (A)  $p + q = 0$  (B)  $p + q = 2n + 1$  (C)  $p + q = 2n$  (D)  $p + q = 2(2n + 1)$

MTE133

2. The general solution of the equation,  $\frac{1 - \sin x + \dots + (-1)^{n-1} \sin^{n-1} x + \dots}{1 + \sin x + \dots + \sin^{n-1} x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$  is  
 (A)  $(-1)^n \pi/3 + n\pi$  (B)  $(-1)^n \pi/6 + n\pi$   
 (C)  $(-1)^{n+1} \pi/6 + n\pi$  (D)  $(-1)^{n-1} \pi/3 + n\pi$

MTE134

## Multiple Type Question

3.  $\cos 15x = \sin 5x$  if  
 (A)  $x = -\frac{\pi}{20} + \frac{n\pi}{5}, n \in I$  (B)  $x = \frac{\pi}{40} + \frac{n\pi}{10}, n \in I$   
 (C)  $x = \frac{3\pi}{20} + \frac{n\pi}{5}, n \in I$  (D)  $x = -\frac{3\pi}{40} + \frac{n\pi}{10}, n \in I$

MTE135

## Subjective Type Question

4. Find general solutions of the following equations:

(a)  $\sin \theta = \frac{1}{2}$  (b)  $\cos\left(\frac{3\theta}{2}\right) = 0$  (c)  $\tan\left(\frac{3\theta}{4}\right) = 0$  (d)  $\cos^2 2\theta = 1$   
 (e)  $\sqrt{3} \sec 2\theta = 2$  (f)  $\operatorname{cosec}\left(\frac{\theta}{2}\right) = -1$

MTE136

5. What are the most general values  $\theta$ , which satisfy the equations :

(i)  $\sin \theta = \frac{1}{\sqrt{2}}$  (ii)  $\tan(x-1) = \sqrt{3}$  (iii)  $\tan \theta = -1$   
 (iv)  $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$  (v)  $2\cot^2 \theta = \operatorname{cosec}^2 \theta$

MTE137

6. Solve

(i)  $\sin 9\theta = \sin \theta$  (ii)  $\cot \theta + \tan \theta = 2 \operatorname{cosec} \theta$  (iii)  $\sin 2\theta = \cos 3\theta$   
 (iv)  $\cot \theta = \tan 8\theta$  (v)  $\cot \theta - \tan \theta = 2$  (vi)  $\operatorname{cosec} \theta = \cot \theta + \sqrt{3}$   
 (vii)  $\tan 2\theta \tan \theta = 1$  (viii)  $\tan \theta + \tan 2\theta + \sqrt{3} \tan \theta \tan 2\theta = \sqrt{3}$

MTE138

MTE139

MTE140

7. Solve the following equations:

(i)  $3\sin x + 2\cos^2 x = 0$

(ii)  $\sec^2 2\alpha = 1 - \tan^2 \alpha$

MTE141

(iii)  $7\cos^2 \theta + 3\sin^2 \theta = 4$

(iv)  $4\cos \theta - 3\sec \theta = \tan \theta$

MTE142

8. Solve  $\cot^2 \theta + 3\operatorname{cosec} \theta + 3 = 0$

MTE143

9. Solve the equation:  $2\sin^2 \theta + \sin^2 2\theta = 2$  for  $\theta \in (-\pi, \pi)$ .

MTE144

10. Solve  $\cos^3 x + \cos^2 x - 4\cos^2 \frac{x}{2} = 0$

MTE145

ANSWER KEY

| Question | 1 | 2 | 3       |
|----------|---|---|---------|
| Answer   | D | B | A,B,C,D |

4. (a)  $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ ; (b)  $\theta = (2n+1)\frac{\pi}{3}, n \in I$ ; (c)  $\theta = \frac{4n\pi}{3}, n \in I$ ;  
 (d)  $\theta = \frac{n\pi}{2}, n \in I$ ; (e)  $\theta = n\pi \pm \frac{\pi}{12}, n \in I$ ; (f)  $\theta = 2n\pi + (-1)^{n+1} \pi, n \in I$
5. (i)  $n\pi + (-1)^n \frac{\pi}{4}, n \in I$ ; (ii)  $n\pi + \frac{\pi}{3} + 1, n \in I$ ; (iii)  $n\pi - \frac{\pi}{4}, n \in I$ ; (iv)  $n\pi + (-1)^n \frac{\pi}{3}, n \in I$ ;  
 (v)  $n\pi \pm \frac{\pi}{4}, n \in I$
6. (i)  $\frac{m\pi}{4}, m \in I$  or  $\frac{(2m+1)\pi}{10}, m \in I$ ; (ii)  $2n\pi \pm \frac{\pi}{3}, n \in I$ ;  
 (iii)  $\left(2n + \frac{1}{2}\right)\frac{\pi}{5}, n \in I$  or  $2n\pi - \frac{\pi}{2}, n \in I$ ; (iv)  $\left(n + \frac{1}{2}\right)\frac{\pi}{9}, n \in I$  (v)  $\left(n + \frac{1}{4}\right)\frac{\pi}{2}, n \in I$ ;  
 (vi)  $2n\pi + \frac{2\pi}{3}, n \in I$ ; (vii)  $n\pi \pm \frac{\pi}{6}$  (viii)  $\left(n + \frac{1}{3}\right)\frac{\pi}{3}, n \in I$
7. (i)  $x = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in I$ ; (ii)  $\alpha = \frac{n\pi}{2}$  or  $\alpha = \frac{k\pi}{2} + \frac{3\pi}{8}, n, k \in I$ ;  
 (iii)  $\theta = n\pi \pm \frac{\pi}{3}, n \in I$ ; (iv)  $\theta = n\pi + (-1)^n \alpha$ ,
8.  $2n\pi - \frac{\pi}{2}, n \in I$  or  $n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in I$  9.  $\theta = \left\{-\frac{\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{2}\right\}$
10.  $(2n+1)\pi, n \in I$



## Trigonometric Equations: RACE-02

## Single Correct Question

1. If sum of all possible values of  $x \in (0, 2\pi)$  satisfying the equation  $2 \cos x \cdot \operatorname{cosec} x - 4 \cos x - \operatorname{cosec} x = -2$ , is  $\frac{k\pi}{4}$ , then  $k$  is equal to
- (A) 9 (B) 12  
(C) 16 (D) 32

MTE146

## Integer Type Question

2. If sum of all solutions of the equation  $6 \sin \frac{x}{2} = \sec \frac{x}{2}$  in  $[-2\pi, 4\pi]$  is  $k\pi$ , then  $k$  is equal to

MTE147

## Subjective Type Question

3. Solve :  $\sin 2x + 5 \sin x + 1 + 5 \cos x = 0$

MTE148

4. Solve

(i)  $\tan^2 \theta - (1 + \sqrt{3}) \tan \theta + \sqrt{3} = 0$

MTE149

(ii)  $4 \cos \theta - 3 \sec \theta = 2 \tan \theta$

MTE150

(iii)  $\tan x \cdot \tan \left( x + \frac{\pi}{3} \right) \cdot \tan \left( x + \frac{2\pi}{3} \right) = \sqrt{3}$

MTE151

5. Solve the following equations :

(a)  $\sin x + \sqrt{2} = \cos x$

(b)  $\operatorname{cosec} \theta = 1 + \cot \theta$

MTE152

6. Solve :  $8 \sin x = \frac{\sqrt{3}}{\cos x} + \frac{1}{\sin x}$

MTE153

7. Solve :

(i)  $\sqrt{3} \sin \theta - \cos \theta = \sqrt{2}$

MTE154

(ii)  $5 \sin \theta + 2 \cos \theta = 5$

MTE155

8. Solve for  $x$ :  $\sin x + \sin 3x + \sin 5x = 0$ . MTE156
9. Solve :  $\cos \theta + \sin \theta = \cos 2\theta + \sin 2\theta$  MTE157
10. Solve :  $4\sin\theta \sin 2\theta \sin 4\theta = \sin 3\theta$ . MTE158

ANSWER KEY

1. (B)    2. 3    3.  $n\pi - \frac{\pi}{4}, n \in I$
4. (i)  $n\pi + \frac{\pi}{3}, n \in I$  or  $n\pi + \frac{\pi}{4}, n \in I$   
 (ii)  $n\pi + (-1)^n \frac{\pi}{10}, n \in I$  or  $n\pi - (-1)^n \frac{3\pi}{10}, n \in I$   
 (iii)  $x = \frac{n\pi}{3} - \frac{\pi}{9}, n \in I$
5. (a)  $x = 2n\pi - \frac{\pi}{4}, n \in I$  (b)  $2m\pi + \frac{\pi}{2}, m \in I$
6.  $x = n\pi + \frac{\pi}{6}, n \in I, x = \frac{n\pi}{2} - \frac{\pi}{12}, n \in I$
7. (i)  $n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{4}, n \in I$ ; (ii)  $2n\pi + \frac{\pi}{2}, n \in I$  or  $2n\pi + 2\alpha$  where  $\alpha = \tan^{-1} \frac{3}{7}, n \in I$
8.  $x = \frac{n\pi}{3}, n \in I$  and  $k\pi \pm \frac{\pi}{3}, k \in I$     9.  $2n\pi, n \in I$  or  $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$
10.  $\theta = n\pi$  or  $\theta = \frac{m\pi}{3} \pm \frac{\pi}{9}; n, m \in I$



## Trigonometric Equations: RACE-03

## Single Correct Question

1. The number of integral values of  $a$  for which the equation  $\cos 2x + a \sin x = 2a - 7$  possesses a solution is

(A) 2 (B) 3  
(C) 4 (D) 5

MTE159

2. If  $x^2 - 4x + 5 - \sin y = 0$ ,  $y \in [0, 2\pi)$ , then -

(A)  $x = 1, y = 0$  (B)  $x = 1, y = \frac{\pi}{2}$   
(C)  $x = 2, y = 0$  (D)  $x = 2, y = \frac{\pi}{2}$

MTE160

3. The number of real solutions of the equation  $\sin(e^x) = 5^x + 5^{-x}$  is -

(A) 0 (B) 1  
(C) 2 (D) infinitely many

MTE161

4. If  $\sin^3 \alpha + \cos^3 \beta + 6 \sin \alpha \cdot \cos \beta = 8$ , where  $\alpha, \beta \in [0, 2\pi]$  then number of ordered pairs  $(\alpha, \beta)$  is equal to

(A) 0 (B) 1  
(C) 2 (D) 3

MTE162

## Subjective Type Question

5. Solve :  $(1 - \sin 2x)(\cos x - \sin x) = 1 - 2 \sin^2 x$ .

MTE163

6. Solve

(i)  $\cos \theta + \sin \theta = \cos 2\theta + \sin 2\theta$

MTE164

(ii)  $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$

MTE165

(iii)  $\sin^2 n\theta - \sin^2(n-1)\theta = \sin^2 \theta$ , where  $n$  is constant and  $n \neq 0, 1$

MTE166

7. Solve  $\sin 3x + \cos 2x = -2$

MTE167

8. Solve  $\sqrt{3\sin 5x - \cos^2 x - 3} = 1 - \sin x$

MTE168

9. If  $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$ ,  $y > 0$ ,  $x \in [0, \pi]$ , then find the least positive value of  $x$  satisfying the given condition.

MTE169

10. Find the values of  $x$  in the interval  $[0, 2\pi]$  for which  $4\sin^2 x - 8\sin x + 3 \leq 0$ .

MTE170

ANSWER KEY

| Question | 1 | 2 | 3 | 4 |
|----------|---|---|---|---|
| Answer   | D | D | A | C |

5.  $2n\pi + \frac{\pi}{2}, n \in I$  or  $2n\pi, n \in I$  or  $n\pi + \frac{\pi}{4}, n \in I$

6. (i)  $2n\pi, n \in I$  or  $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

(ii)  $x = (2n + 1)\frac{\pi}{4}, n \in I$  or  $x = (2n + 1)\frac{\pi}{2}, n \in I$  or  $x = n\pi \pm \frac{\pi}{6}, n \in I$

(iii)  $m\pi, m \in I$  or  $\frac{m\pi}{n-1}, m \in I$  or  $\left(m + \frac{1}{2}\right)\frac{\pi}{n}, m \in I$

7.  $(4p-3)\frac{\pi}{2}, p \in I$       8.  $2m\pi + \frac{\pi}{2}, m \in I$       9.  $x = \frac{\pi}{4}$       10.  $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$